Chapter 7: Standards in SQA management

Reference

This chapter refers from the book:

Mastering Software Quality Assurance: Best Practices, Tools and Techniques for Software Developers

Introduction to Software Testing

Outline

7.1. The scope of quality management standards

7.2. SQA in ISO standard

7.3. SQA in CMM, CMMI standard

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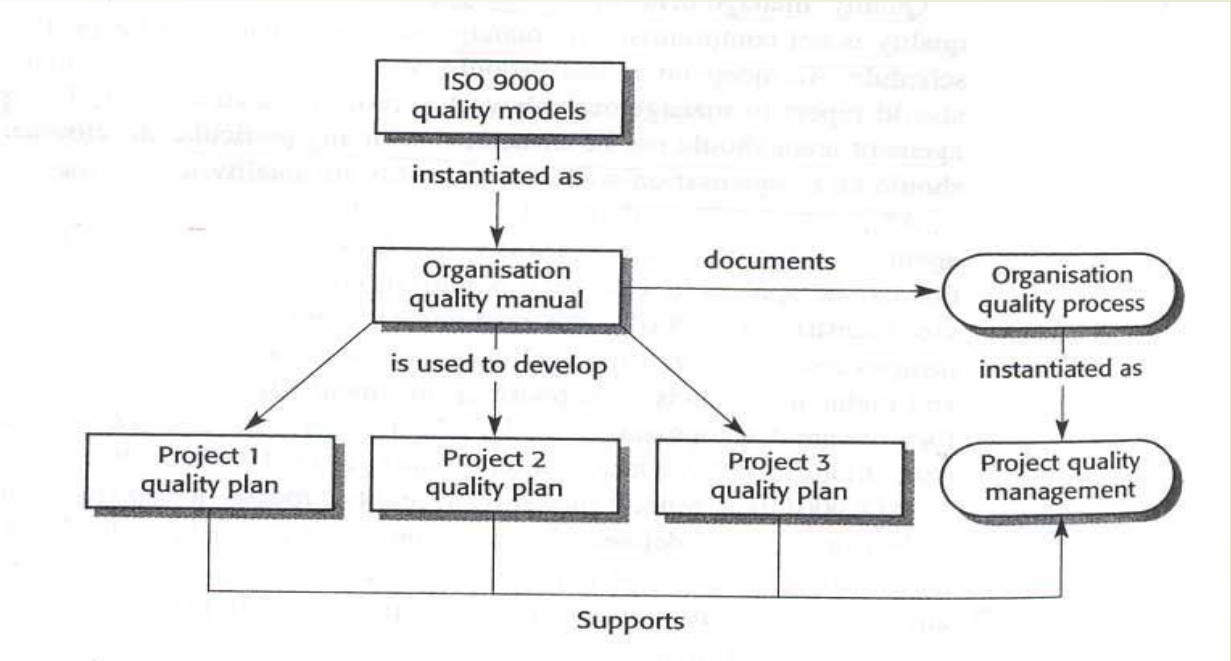
Why are Standards Important?

- Standards provide encapsulation of best, or at least most appropriate, practice
- Standards provide a framework around which the quality assurance process may be implemented
- Standards assist in continuity of work when it's carried out by different people throughout the software product lifecycle

Standards should not be avoided. If they are too extensive for the task at hand, then they should be tailored.

A Simplistic approach

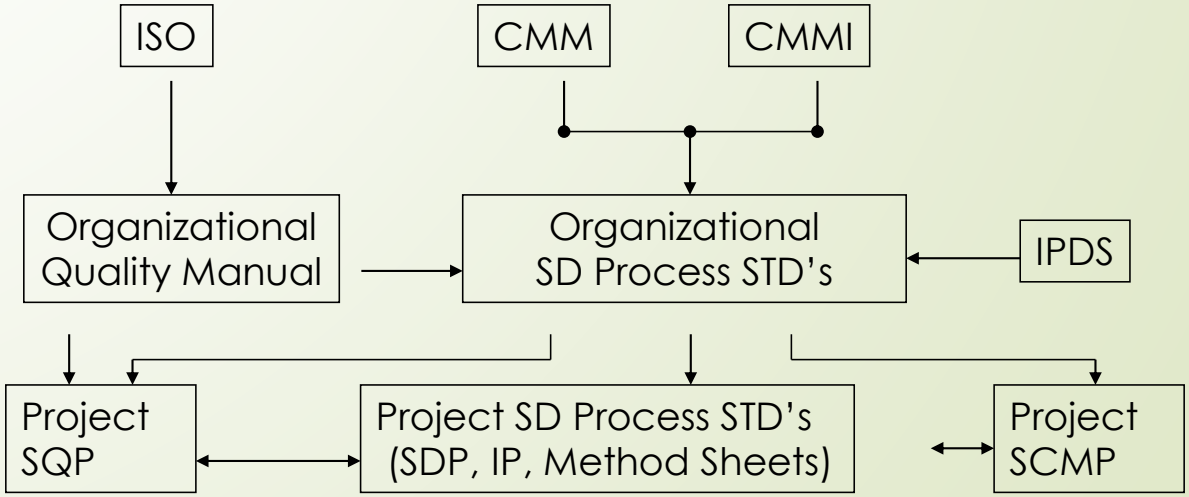


In most mature organizations:

- ISO is not the only source of SDS
- Process and Product standards are derived independently
- Product standards are not created by SQA

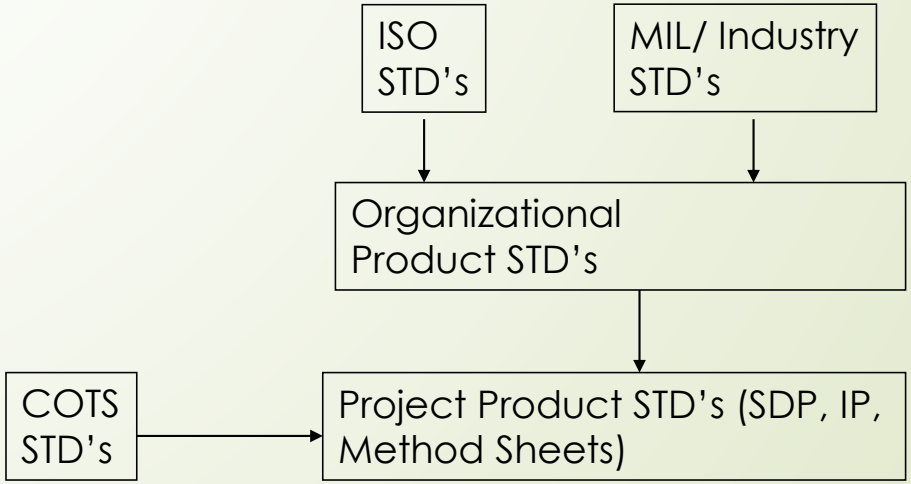
Process Standards

Process Standards – standards that define the process which should be followed during software development



Product Standards

Product Standards – standards that apply to software product being developed



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ISO 9000

- ISO (International Standards Organization):
  - a consortium of 63 countries established to formulate and foster standardization.
- ISO published its 9000 series of standards in 1987.
- A set of guidelines for the production process.
  - not directly concerned about the product it self.
  - a series of three standards: ISO 9001, ISO 9002, and ISO 9003.
- Based on the premise: if a proper process is followed for production, good quality products are bound to follow.

The ISO 9000:2000 Software Quality Standard

- The international organization ISO has developed a series of standards for quality assurance and quality management, collectively known as the ISO 9000.
- The ISO 9000 standards are reviewed and updated once every 5-8 years. The standards released in the year 2000 are known as ISO 9000:2000.
- There are three components of the ISO 9000:2000 standard.
  - ISO 9000: Fundamentals and vocabulary
  - ISO 9001: Requirements
  - ISO 9004: Guidelines for performance improvements
- Note: ISO 9002 and ISO 9003 were parts of ISO 9000:1994, but these are no more parts of ISO 9000:2000.

ISO 9000 for Software Industry

- ISO 9000 is a generic standard:
  - applicable to many industries,
    - starting from a steel manufacturing industry to a service rendering company.
- Many clauses of ISO 9000 documents:
  - use generic terminologies
  - very difficult to interpret them in the context of software organizations.
- Very difficult to interpret many clauses for software industry:
  - software development is radically different from development of other products.

Software vs. other industries Cont…

- Software is intangible
  - therefore difficult to control.
    - It is difficult to control anything that we cannot see and feel.
  - In contrast, in a car manufacturing unit
  - Software project management is an altogether different ball game.
- During software development: the only raw material consumed is data.
- For any other product development: Lot of raw materials consumed..
- ISO 9000 standards have many clauses corresponding to raw material control .

  -> not relevant to software organizations.

The ISO 9000:2000 Software Quality Standard

ISO 9000:2000 Fundamentals:

- This is based on *eight principles*.
  - Principle 1: Customer focus
  - Principle 2: Leadership
  - Principle 3: Involvement of people
  - Principle 4: Process approach
  - Principle 5: System approach to management
  - Principle 6: Continual improvement
  - Principle 7: Factual approach to decision making
  - Principle 8: Mutually beneficial supplier relationships

The ISO 9000:2000 Software Quality Standard

ISO 9001:2000 Requirements

- The five major parts of this document are as follows.
  - Part 4. Systemic requirements
  - Part 5. Management requirements
  - Part 6. Resource requirements
  - Part 7. Realization requirements
  - Part 8. Remedial requirements

The ISO 9000:2000 Software Quality Standard

ISO 9001:2000 Requirements

- Part 4. Systemic requirements (partial)
  - Document the organizational policies and goals.
  - Document all quality processes and their interrelationship.
  - Implement a mechanism to approve documents before are distributed.
- Part 5: Management requirements (partial)
  - Generate an awareness for quality to meet a variety of requirements, such as customer, regulatory, and statutory.
  - Focus on customers by identifying and meeting their requirements in order to satisfy them.
  - Develop a quality policy to meet the customer's needs.
  - Clearly define individual responsibilities and authorities concerning the implementation of quality policies.

The ISO 9000:2000 Software Quality Standard

ISO 9001:2000 Requirements

- Part 6. Resource requirements (partial)
  - Identify and provide resources required to support the organizational quality policy to realize the quality objectives.
  - Allocate quality personnel resource to projects.
  - a mechanism to enhance the quality level of personnel.
- Part 7: Realization requirements (partial)
  - Develop a plan to realize a product from its requirements.
  - Gather their requirements. Classify those requirements.
  - Review the requirements.
  - Follow a defined purchasing process by evaluating potential suppliers based on a number of factors, such as ability to meet requirements and price.

The ISO 9000:2000 Software Quality Standard

ISO 9001:2000 Requirements

- Part 8. Remedial requirements (partial)
  - Measure and track the customer's satisfaction level.
  - Perform internal audit.
    - Example: Find out whether or not personnel with adequate education, experience, and skill have been assigned to a project.
  - Monitor processes by using a set of key performance indicators.

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Capability Maturity Model (CMM)

The Software engineering Institute – Capability Model (SEI-CMM) is a model for judging the following:

- Judging the maturity of the software processes of an organization.
- Identifying the key practices that are required to increase the maturity of these processes.
- Describes the principles and practices underlying software process maturity and is intended to help software organization improve the maturity of their software processes in terms of an evolutionary path from ad hoc chaotic processes to mature software process

Capability Maturity Model (CMM)

The CMM is organized into five maturity levels

Level 1 : Initial

- The software process is characterized as ad hoc, few processes are defined and success depends on individual efforts.
- This period is chaotic without any procedure and process established for software development and testing.

Capability Maturity Model (CMM)

Level 2 : Repreatable

- Track cost, schedule, and functionality .
- During this phase, measures and metrics will be reviewed to include percentage compliance with various processes, percentage of allocated requirements delivered, number of changes to requirements, number of changes to project plan, variance between estimated and actual size of deliverables.
- The following are the key process activities during Level 2:
  - Software configuration management
  - Software quality assurance
  - Software subcontract management
  - Software project tracking and oversight
  - Software project planning
  - Requirement management

Capability Maturity Model (CMM)

Level 3: Defiened

- The software process for management and engineering activities is documented, standardized and integrated into a standard software process for the organization.
- All projects use an approved version of the organization standard software process for developing and maintaining software.
- The following are the key process activities during Level 3:
  - Examine reviews
  - Intergroup coordination
  - Software program engineering
  - Integrated software management
  - Training Program
  - Organization process definition
  - Organization process focus

Capability Maturity Model (CMM)

Level 4: Managed

- Detailed measures of the software process and product quality are collected and both are understood and controlled.
- This phase denotes that the processes are well defined and proficiently managed.
- The quality standard are on an upswing.
- With sound quality process in place the organization is better equipped to meet customer expectations of high quality/ high performance software at reasonable cost and commitment deliveries .

Capability Maturity Model (CMM)

Level 5: Optimizing

- Continues process improvement is enabled by quantitative feedback from the process and from piloting new idea and technologies.
- Continuous emphasis on process improvement and defect reduction avoid process stagnancy and ensure continual improvement translating into improved productivity, tracing requirements across each development phase improves the completeness of software, reduce rework, and simplify maintenance. Verification and validation activities are planned and executed to reduce defect leakage. Customers have access to the project plane, receive regular status reports and their feedback is sought and used for process tuning.

What is a CMMi?

- A Capability Maturity Model (CMMi) is a reference model of mature practices in a specified discipline, used to improve **PROCESS** at work
- The results of adopting CMMi is a much better product or process quality.
- Before we focus on CMMi we need to understand the meaning of a **PROCESS**

so What is a PROCESS ?

What is a process

- A process is a series of actions or steps taken in order to achieve a particular end in the form of a product or service
- We may not realize it, but processes are everywhere and in every aspect of our leisure and work. A few examples of processes might include:
  - Preparing breakfast
  - Placing an order
  - Developing a budget
  - Writing a computer program
  - Obtaining application requirements
  - And so on

Process Improvement

- The quality of a system is highly influenced by the quality of the process used to acquire, develop, and maintain it.
- even our finest people can't perform at their best when the process is not understood or operating at its best."
- Everyone realizes the importance of having a motivated, quality work force and the latest technology, but even the finest people can't perform at their best when the process is not understood or operating at its best
- This premise implies a focus on processes as well as on products.

CMMI for Process Improvement

The aim of CMMi is to improve processes so they can be performed in the best manner with least cost

Use CMMI in process improvement activities as a

- collection of best practices
- framework for organizing and prioritizing activities
- support for the coordination of multi-disciplined activities that might be required to successfully build a product
- means to emphasize the alignment of the process improvement objectives with organizational business objectives

Ad Hoc Processes (Not using CMMi)

Processes are ad hoc and improvised by practitioners and their management

Process describes are not rigorously followed or enforced Performance is highly dependent on current practitioners

Understanding of the current status of a project is limited

Immature processes result in fighting fires:

- There is no time to improve – instead,

  practitioners are constantly reacting
- Firefighters get burned
- Embers might rekindle later

Improved Processes (Using CMMi)

- Process descriptions are consistent with the way workactually is done
- They are defined, documented and continuously improved
- Processes are supported visibly by management and others
- They are well controlled – process fidelity is evaluated and enforced
- There is constructive use of product and process enforced
- There is constructive use of product and process measurement
- Technology is introduced in a disciplined manner

… So what is CMMI?

- In the same way, high-quality software organizations are different from low-quality organizations.
- CMMI tries to capture and describe these differences.
- CMMI strives to create software development organizations that are "mature", or more mature than before applying CMMI.

How CMMI Helps?

CMMI provides guidance for improving an organization's processes and ability to manage the development, acquisition and maintenance of products or services.

CMMI places proven approaches into a structure that helps an organization:

- appraise its organizational maturity or process area capability
- establish priorities for improvement
- implement these improvements

Summary of levels

- Level 1 – Initial. Anything at all. Ad-hoc and chaotic. Will have some successes, but will also have failures and badly missed deadlines.

- Level 2 – Repeatable. SW processes are defined, documented, practiced, and people are trained in them. Groups across an organization may use different processes.

Summary of levels

- Level 3 – Defined. SW processes are consistent and known across the whole organization.
- Level 4 – Managed. SW processes and results are measured quantitatively, and processes are evaluated with this data.
- Level 5 – Optimizing. Continuous process improvement. Experimenting with new methods and technologies. Change processes when find something that works better.

Level 1 – Initial

- Team tackles projects in different ways each time
- Can have strong successes, but may not repeat
- Some time/cost estimates are accurate, many far off
- Success comes from smart people doing the right things
- Hard to recover from good people leaving
- Frequent crises and "firefighting." (Many believe this is standard for SW development. CMM says NO.)
- Most SW development organizations are Level 1.

Level 2 – Repeatable

Key areas

- Requirements management
- Software project planning
- Project tracking and oversight
- Subcontracts management
- Quality assurance
- Configuration management

Level 3 – Defined

Key areas. Level 2, plus…

- Organization-wide process focus
- Organization-wide process definition
- Training program in above
- Integrated software management (above applied per project)
- Software product engineering (coding, etc.)
- Inter-group coordination
- Peer reviews

Level 4 – Managed

Key areas. Level 3, plus…

- Quantitative process management (data gathering)
- Quality management (data-driven quality improvement)

Level 5 – Optimizing

Key areas. Level 4, plus…

- Defect prevention
- Technology change management (bring in new methods)
- Process change management (improve processes)

- The optimizing level (Level 5) is not the destination of process management.
- The destination is better products for a better price: economic survival
- The optimizing level is a foundation for building an ever-improving capability.